POC SUMMARY · BRIEF 12 · CLEAN ENERGY SOLUTIONS

Relocatable clean power for a hospital on the move

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Focus region Enugu, Nigeria | **Worked example** National Orthopaedic Hospital Enugu | **Team solution** GridRescue AI

THE PROBLEM IN BRIEF 12

The National Orthopaedic Hospital Enugu receives about six hours of grid power a day against about 1,550 kW of connected capacity, and the unstable supply keeps damaging sensitive equipment [1]. When the grid drops mid-procedure, the theatre waits while staff start the diesel generator [11]. Diesel averaged ₦3,277 a litre in Nigeria in May 2026, up 86 % in a year [4], and federal teaching hospitals have seen electricity bills rise three to four times since the 2024 Band A tariff increase [3]. Enugu now runs its own electricity market: the state regulator cut Band A tariffs to ₦160/kWh and requires published feeder supply hours [5]. The hospital plans to move to a new building in about five years, and federal asset rules prevent it from simply disposing of solar fixed to today's buildings [1]. Nigeria has lost about 15,000–16,000 doctors to emigration in five years, and working conditions, including power, drive the exits [6].

Precedents point the way: a 1.5 MW solar hybrid at the sister orthopaedic hospital in Dala, Kano [2]; the EU and Enugu State's ₦4.3 bn solar programme for 25 health facilities, with local businesses and telecom towers as anchor customers [10]; and a UNHCR camp hospital in Tanzania that runs on 77 kWp of solar with 320 kWh of containerised battery storage, proof that a system can be built to move [8].

WHAT WE ARE SOLVING FOR: THE THREE A'S

ACCESS	AFFORDABILITY	AVAILABILITY
Bring reliable power to hospitals and communities still living without it.	Drive down the cost of clean energy so it is within reach for the hospital board.	Build resilient systems that keep energy flowing anytime, anywhere, including during a move.
Today: about 6 h of grid a day against 1,550 kW of need [1]; the state publishes feeder supply hours [5].	Today: diesel at ₦3,277/L, up 86 % in a year [4]; grid at ₦160/kWh in Enugu [5]; bills three to four times higher since 2024 [3].	Today: the grid drops without warning and switchovers break equipment [1]; the hospital moves in about five years [1].
We help: any site is read from open data in seconds; for Enugu the engine finds 9 substations within 50 km and 5.48 peak-sun hours [13].	We help: ₦0 upfront; a ₦69.1 M monthly service fee replaces ₦299.2 M a month of modelled diesel (plan capex ₦3.85 bn); payback 1.6 years [13].	We help: 4 of 4 stress tests pass; 100 % survival of any outage up to 24 h, whatever hour it starts; 88 % of the investment moves with the hospital [13].

WHAT WAS BUILT · THE PROOF OF CONCEPT

GridRescue AI is a working web application. Everything below runs today in the demo, in this order, for any pin on the map; Enugu is the worked example. Every result exports as GeoJSON for QGIS, and a data-sources tab lists where each number came from.

STEP	WHAT THE DEMO SHOWS
1 Grid mapper	Pin the hospital. A sun lens reads the site from four public sources: NASA POWER, PVGIS, PVWatts and the Global Solar Atlas [15]–[18]. The same lens reads wind and water so the board sees every local resource, not only solar.
2 "The hospital is moving" lens	Areas within 12 km of the current site are scored on grid access (OpenStreetMap substations and lines [19]) and distance. The user picks a candidate area and locks the new site.
3 Grid power at the site	Substations within 50 km and 100 km, published supply hours and class, and the published tariff (Enugu Band A ₦160/kWh [5]) are shown for the locked site.
4 Relocation plan	The five starter questions of Brief 12 [11] are answered on one screen: which loads must never lose power, how the system moves, who should own it, whether diesel savings repay it, and which funders have backed similar projects.

STEP	WHAT THE DEMO SHOWS
5 Two sites, one check	Current and new site side by side: grid price, a grid-first feeder check (modelled from published hours) and sun confirmed by the four sources.
6 Energy type	LFP (lithium iron phosphate) containers, flow batteries and hydrogen are each simulated hour by hour on the site's sun, grid hours and load; if storage must wait, a grid-first UPS fallback.
7 Impact	Three A's, diesel litres and net CO_2 avoided, ESG lines by GHG Protocol scope [14]; carbon credits: potential, valued at the buyer's price only after verification and issuance.
8 3D simulation	Hospital: ground-mounted solar, 5 containers (3 of them battery), 500 kVA UPS, generator last, in a 24 h national collapse. Rail night crew: a battery trailer replaces the diesel generator, 45 dB vs 68 dB [13].
9 ESG report	For the selected site, an ESG report that saves as PDF: Scope 1 (diesel) and Scope 2 (grid) under the GHG Protocol [14], E · S · G lines, the Three A's, and the sources behind every figure.

HOW THE GRIDRESCUE ENGINE WORKS

The engine is a deterministic simulation that runs hour by hour on a representative design day for each outage case. It takes the sun at the site, the published grid hours and the hospital's load, and runs each storage family through the same day: charge when sun or grid is there, discharge when not, keep the theatres on at all times, start the generator only when everything else is used up. The cases are four stress tests (12 h evening outage, 24 h national collapse, worst solar month, 18 h outage) and a survival check in the style of NREL's REopt: an outage is started at each of the 24 hours of the day and must be carried for up to 24 h. Every step is shown on screen so an engineer can check it.

WORKED EXAMPLE: NATIONAL ORTHOPAEDIC HOSPITAL ENUGU

Engine results for the worked example; to be confirmed with the hospital's load data (partner data not yet received) [13]. Site read: 350 beds, 1,550 kW connected, grid 6 h/day (class D), 9 substations within 50 km, 5.48 peak-sun hours, tariff ₦160/kWh.

STARTER QUESTION [11]	ENGINE ANSWER FOR ENUGU [13]
Which loads must never lose power?	Critical load 303 kW (surgical theatres 91 kW) on a 500 kVA online UPS with 0 ms switchover, so surgery never feels a switchover; protected load 641 kW.
What should be installed?	1,670 kWp ground-mounted solar; 5,020 kWh LFP storage (8.7 h of protected load) in 5 containers (3 of them battery, plus the inverter skid and the UPS e-house); 500 kVA UPS; the existing 1,000 kVA generator as the last resort. Capex ₦3.85 bn.
Will the theatres stay on?	4 of 4 outage stress tests pass; outage survival is 100 % for outages up to 24 h, whatever hour they start. 4,992 theatre-hours a year (40 %) fall when the grid is off today, 0 with the system; the UPS removes about 1,460 generator switchovers a year (modelled).
Can diesel savings repay it, and how fast?	Modelled diesel today: about 1,042,790 L a year, ₦299.2 M a month for the protected load (the real bill is partner data still owed). Energy-as-a-service: ₦0 upfront, fee ₦69.1 M a month; it saves money from month one whenever the diesel bill is above ₦117.9 M a month. Payback from diesel savings 1.6 years; 866,678 L of diesel avoided a year (83 %).
How does it move in five years?	To lens area #1, 4 km north (grid-access score 98/100), in about three weeks for ₦678 M (17.6 % of the investment); 88 % of the investment is reused, while solar on today's roof would strand ₦1,510 M [1].
Who owns it, and who funds it?	A private energy-service company owns, insures and moves the kit, with concessional debt. Funders: the Rural Electrification Agency's National Public Sector Solarisation Initiative (precedent: the 1.5 MW solar hybrid at the orthopaedic hospital in Dala, Kano [2]); the European Union and Enugu State's Solar for Health (₦4.3 bn, 25 health facilities [10]); development-bank financing for renewable electrification of health facilities (US$1.7 bn committed [7]).

STARTER QUESTION [11]	ENGINE ANSWER FOR ENUGU [13]
Which storage family?	The energy-type comparison is a different design (solar plus a bigger battery, no generator): LFP ₦4.65 bn versus hydrogen ₦8.18 bn, almost twice, both keeping theatres at 100 %. The plan above keeps the generator as the last resort: ₦3.85 bn.

5- AND 10-YEAR COST MODEL (BRIEF 12 DELIVERABLE)

COST OF POWER FOR THE PROTECTED LOAD [13]	5 YEARS	10 YEARS
Diesel, as modelled today	₦18.8 bn	₦37.6 bn
Grid only (reference: a 24 h grid, which the hospital does not have today [1])	₦3.5 bn	₦7.0 bn
Solar + storage service (this plan)	**₦7.9 bn**	**₦15.9 bn**

Saving versus diesel over 10 years: ₦21.8 bn (figures rounded).

WHY THE BOARD NEEDS THE ESG REPORT

Nigeria's Financial Reporting Council is rolling out the ISSB standards (IFRS S1 and S2), which ask for Scope 1 and Scope 2 emissions [20], and calls climate and sustainability governance a core board duty [21]. Hospitals use about 2.5 times the energy of an equivalent commercial building [22]; Evercare Hospital Lekki became Nigeria's first EDGE-certified hospital [23]. GridRescue produces the site's ESG report before purchase (Scope 1 and 2, E · S · G lines, carbon credits valued only after verification). For Enugu: 2,246 t CO_2 avoided a year, net = 2,323 t from diesel no longer burnt (Scope 1) minus 77 t because a little more grid electricity is bought to charge the batteries for the night (Scope 2, netted off) [14]; 22,460 t over 10 years [13].

OUTAGE TYPES THE ENGINE COVERS, AND CAN EXTEND TO

STATUS	OUTAGE TYPES, BY LEVEL
Built	Site: 12 h evening outage (design case), worst solar month, 18 h outage. Grid: 24 h national grid collapse. Survival check: an outage started at each of the 24 hours, up to 24 h long.
Extension	Grid: feeder or substation trip, load shedding and scheduled band cuts, frequency collapse. Location: one feeder, district or state market; storm, flood, heat wave. Grid / site: cyber-security outages that cut power or controls.

ONE ENGINE, MANY USE CASES

The same site read and hourly simulation serve any load that runs on diesel today, with cleaner, quieter power and less switchover damage [1].

USE CASE	STATUS	WHAT THE ENGINE DOES
Hospital (Brief 12)	Built	Critical-load list, relocatable solar and containerised storage, move plan, 5- and 10-year cost model, funder case.
Night rail maintenance crew	Built	A battery trailer sized for the shift replaces the diesel generator: 45 dB instead of 68 dB [13].
Clinic cold storage (Brief 01)	Extension	Solar fridges fail through batteries and controllers with no repair system [12]; the engine can size one for the worst solar month.
Surplus solar to neighbours	Extension	Where state rules allow, surplus daytime power goes to anchor customers such as businesses and telecom towers, as in the EU–Enugu programme [10].
EV charging, telecom towers, feeder planning	Extension	EV charging soaks up surplus solar; towers are the same diesel-to-battery case on a smaller load; utilities see which feeders need storage first.

HOW WE MEET THE JUDGING CRITERIA [11]

JUDGED ON	OUR EVIDENCE
Impact · 25 %	4,992 theatre-hours a year (40 %) fall when the grid is off today, 0 with the system; 866,678 L of diesel (83 %) and 2,246 t CO_2 (net) avoided a year; the UPS removes about 1,460 generator switchovers a year (modelled) [13].
Innovation · 25 %	A relocatable design chosen by hour-by-hour simulation of every storage family, plus a lens that scores the new site and a move plan with 88 % of the investment reused.
Sustainability · 25 %	Containers and ground mounts move with the hospital (₦678 M, 88 % reused); 22,460 t CO_2 and ₦21.8 bn saved over 10 years; maintainable locally, with the existing generator as the last resort.
Feasibility · 15 %	Sized on published grid hours, tariff and four solar sources; 4 of 4 stress tests pass and any outage up to 24 h is survived; to be re-run on the hospital's own load data.
Scalability · 10 %	The data (sun, weather, OpenStreetMap grid) is global; a new country needs only its tariff table. Vendor-neutral: works with any supplier, and we sell no hardware.

OUR TEAM SUBMITS A demo video (POC walkthrough, under 5 minutes), a pitch video (under 5 minutes), a 5-slide pitch deck, this POC summary and a disclosure statement.

REFERENCES

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Figures marked [13] are engine results for the worked example; all other figures are from the cited sources. Final sizing will be confirmed with the hospital's load data.